Learning to Select Actions for Complex Hand Tasks

A Comparative Study of Visuomotor Policies and Latent World Models

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ABSTRACT

A latent world model predicts how a scene will evolve in a compact learned representation, so it can judge a proposed action by its predicted consequences. This paper asks whether a frozen world model, used only as the final decision stage over candidate hand motions that a visuomotor policy already proposes, improves the motion that is selected. A controlled offline benchmark of 7,607 held-out EgoDex windows compares the policy alone, world-model scoring with the policy's preferred candidate retained, and scoring with that candidate excluded. Two policies and five world-model families are compared under matched local training exposure, with two mechanism controls. Diffusion Policy candidates re-ranked by V-JEPA2-AC reduce translation error by 5.98% and rotation error by 0.96 degrees, pooled across three training seeds and without retraining, recovering about 36% of the error a perfect selector could remove from the same five candidates. The other families give null, mixed, or unvalidated results, each with a likely explanation or limitation. A secondary reproduction on CALVIN shows that finetuning a policy inside a world model moves task success by 14 to 40 percentage points, roughly an order of magnitude more than external selection. Across the combinations tested, guidance helps when the candidate pool is diverse, the scorer receives meaningful action-conditioned information, and the candidates lie within the scorer's training support.

Keywords world models, imitation learning, diffusion policy, action chunking, candidate selection

1. INTRODUCTION

Imitation learning acquires manipulation behaviour directly from demonstrations, which suits tasks whose useful motion patterns are hard to encode by hand. Behaviour cloning maps observations to demonstrated actions, but small errors compound across a predicted sequence, so agreement with a demonstration at each step does not guarantee agreement over the whole motion [1].

Visuomotor policies connect observations to future motion. Two families span the main design choices in current practice. Action Chunking with Transformers (ACT) regresses an extended action chunk in a single deterministic pass [2], while Diffusion Policy (DP) samples chunks from a learned generative process [3]. A policy that can propose several chunks still needs a criterion for choosing between them. For the policies studied here that choice is a fixed rule, such as taking the policy's own preferred sample, which cannot look ahead to see whether a motion leads to a plausible state. This study instead uses a latent world model as the selector, since it predicts how a state may evolve and can score a candidate by its predicted outcome [4].

This paper studies that pairing in its most conservative form. The world model never generates motion. Every candidate is proposed by the policy, and the world model acts only at the moment of selection. The research question is whether a frozen world model, acting as the final decision stage over candidates a policy already proposes, improves the selected motion chunk. Three objectives follow.

1. Build a controlled offline benchmark that separates the policy's preference from the world model's.
2. Evaluate five world-model families under matched local training exposure.
3. Determine when selection helps, and diagnose each case where it does not.

The contributions are as follows. First, a three-branch evaluation protocol with frozen candidate pools, blocked access to the future goal, and an oracle ceiling that bounds what any selector could gain. Second, one validated positive result, DP candidates scored by V-JEPA2-AC, together with a likely explanation for each of the four other routes that did not reproduce it. Third, a secondary reproduction of DiWA on CALVIN [5], which supports a three-condition reading of when external scoring helps and when weight-level finetuning is the stronger mechanism.

The paper is organised as follows. Section 2 reviews related work. Section 3 describes the benchmark and models. Section 4 reports results. Section 5 discusses interpretation and limitations, and Section 6 concludes.

2. RELATED WORK

2.1 Chunked Visuomotor Policies

ACT trains a conditional variational autoencoder (CVAE) [6, 7] that predicts a chunk of future actions, with a latent variable intended to capture variation between demonstrations [2]. At inference the latent is normally fixed, so the policy offers one deterministic answer. DP instead denoises Gaussian noise into an action chunk conditioned on the observation, and drawing several noise seeds gives several distinct chunks [3]. DP therefore supplies a natural candidate pool, and ACT, as Section 3.2 shows, may not.

2.2 Latent World Models

Latent world models predict in a learned representation space and not in pixels. The families considered here differ mainly in what they are built on and how heavy they are.

- **V-JEPA2-AC.** Builds on joint-embedding predictive architectures. V-JEPA2 extends the idea to video, pretrained on a far larger corpus, and the action-conditioned variant adds a transition head that predicts the next latent state from an action [8].
- **DexWM.** Applies the same action-conditioning idea to

dexterous manipulation, with a DINOv2 encoder and a conditional diffusion transformer predictor. It is the heavier and costlier route [9].

- **LeWM.** Takes the opposite route and compacts the model, training a small encoder and predictor end to end from task data [10].
- **DINO-WM and JEPA-WM.** Released checkpoints that expect a seven-dimensional DROID-style action interface [11, 12].
- **PointWorld.** Reasons over point clouds with arm kinematics, which does not map directly onto the continuous wrist transforms of human demonstration [13].

2.3 World Models in the Policy Loop

Several systems use a world model inside planning or training. TD-MPC2 and DreamerV3 plan or learn behaviour in a learned model [14, 15]. DiWA pairs a diffusion policy with a frozen world model and a reward classifier, then finetunes the policy with PPO on imagined rollouts [5]. The world model there is a training signal, so its judgement is written into the policy weights. The present study sits at the opposite end, with an external, inference-time scorer that cannot change the policy. The gap addressed here is a controlled comparison of that external role across several world-model families, with an explicit bound on what selection can achieve.

2.4 Offline and Closed-Loop Evaluation

World models are often judged by task success under closed-loop execution, which is the standard DexWM and DiWA report [5, 9]. Offline trajectory matching is a different and weaker modality. It measures how closely a selected motion matches an expert demonstration, which is useful for imitation fidelity but does not establish that a motion would succeed physically. This study uses the offline modality for its main results because it permits a large, frozen, repeatable test set, and it adds a closed-loop check through the CALVIN reproduction.

3. METHODOLOGY

3.1 Benchmark and Data

Experiments use EgoDex, a large egocentric video dataset of human manipulation [16]. Table 1 summarises the setup.

Table 1. Benchmark setup.

Test set	7,607 windows, 3,229 episodes, 111 tasks
Window	16 observed frames, 16-frame target
Action	18-D bimanual wrist motion (SE(3))
Policies	ACT, Diffusion Policy (DP-T)
World models	V-JEPA2-AC, DexWM, LeWM, DINO-WM, JEPA-WM, PointWorld proxy
Pool size	$K = 5$ candidates
Exposure	100,000 items for every locally trained model
Seeds	42, 43, 44, pooled

The candidate pool was settled at five after a sweep from $K = 2$ to $K = 20$, which left the ranking of scorers unchanged and the share of the oracle ceiling recovered (capture) near 40%, so larger pools added headroom without better selection. Full sweeps are in the accompanying dissertation. Every locally trained model received 100,000 training items. In the convergence study only two of the five primary models, ACT and LeWM, had converged by this point, so the figure is a matched budget and not a convergence claim. Training was repeated at three seeds (42, 43, and 44) as a robustness check, although three is a small cohort, and DexWM was run at one seed because of its cost. All three settings are compute-bounded.

3.2 Candidate Pools

Both policies share a ResNet-18 observation encoder, and each must supply five candidate chunks per window, with candidate 0 as the policy's own preference.

Diffusion Policy runs five independent DDIM chains [17] from different Gaussian noise under the same conditioning, which return five distinct chunks. Candidate 0 is the chain with the lowest final denoising residual, since the published architecture does not rank its own samples.

ACT varies its output by sampling a latent vector z. In the default configuration the latent suffered posterior collapse [18] in one of three seeds, so the five samples were identical chunks, while the other two seeds were unaffected. Four retraining variants, with a larger KL weight, cyclical annealing, observation dropout, and free bits [19], did not remove the collapse, which appeared in 7 of 12 checkpoints. The observations are informative enough that the decoder minimises its loss without using z. Whenever collapse is detected, a cross-entropy planner (CEMP) [20], inspired by the policy-guided planning in TD-MPC2 [14], perturbs ACT's single output and keeps the perturbations the world model scores highest. The world model therefore appears twice, once to build the pool and again to select from it (Figure 1), which introduces a bias. ACT results measure the planner and world model as one system, and DP, whose pool involves no world model, carries the primary claim.

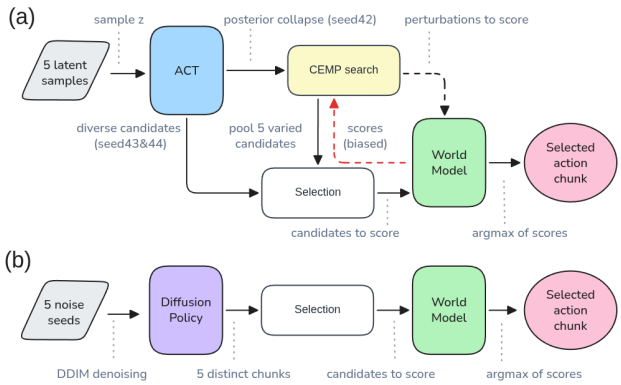


Figure 1. How the candidate pool is built. (a) ACT's sampled latent collapses to one chunk, so a search queries the world model, which later makes the final selection, biasing the choice toward what the scorer already prefers. (b) DP's candidates are generated without the world model.

3.3 Three Branches

Figure 2 shows the evaluation structure.

- **Branch 1 (B1)** is the policy alone and defines the baseline.
- **Branch 2 (B2)** lets the policy generate five candidates and uses the frozen world model to score and rank them, selecting the argmax, so the policy's own choice survives only if the scorer ranks it first.
- **Branch 3 (B3)** keeps the same candidate generation

and world-model selection as B2, but candidate 0, the policy's preference, is omitted.

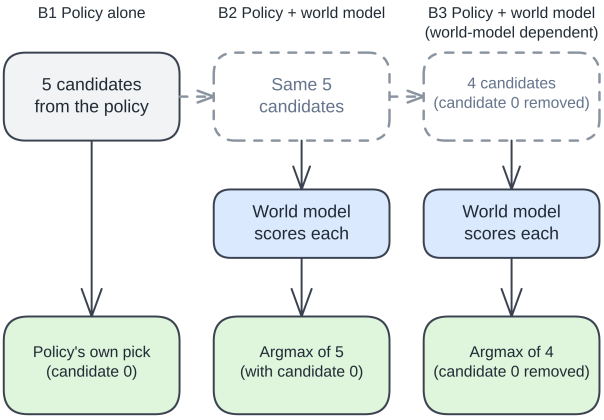


Figure 2. Three-branch evaluation. B2 and B3 reuse the pool generated for B1 (dashed arrows). The world model never proposes motion.

All branches share the same policy weights, windows, and candidate pool, so a difference between them is attributable to the selection rule. Reading B2 and B3 together separates a scorer that merely avoids the policy's bad choices from one that finds better candidates elsewhere in the pool. B2 is expected to outperform B3, since B3 has less to choose from and candidate 0 is not a bad choice when it is the policy's favourite. B3 removes it deliberately, to test whether the world model can still select a valid candidate outside the policy's preference, which shows its choices are justified and not borrowed from the policy.

3.4 Scorers and Fairness

V-JEPA2-AC uses a frozen ViT-Large backbone pretrained on over one million hours of video, with an action-conditioned head trained locally. Scoring predicts forward in the world model's latent space from the current observation and measures how well each candidate action is supported by that prediction. Two controls isolate its mechanism. V-JEPA2 without action conditioning keeps the same backbone but removes the action input, and a PointWorld proxy reuses the visual pathway with weakened action conditioning. DexWM, LeWM, DINO-WM, and JEPA-WM complete the set. Locally trained models share the matched exposure above, but models with externally pretrained backbones still carry training this budget does not offset, so deltas are read model by model and not as an architecture ranking.

3.5 Metrics and Statistics

Translation error is the root-mean-square error per axis and rotation error is the mean geodesic angle in degrees, each averaged over the 16-step chunk. Results are reported as the change relative to candidate 0, with negative values an improvement. Confidence intervals are 95% episode-clustered bootstrap intervals pooled across seeds. The oracle ceiling is the reduction a selector with access to ground truth would obtain from the same pool, and capture is the share of that ceiling a scorer recovers. A ranking-validity gate checks that a scorer's output predicts which candidate is actually better, measured as the per-window Spearman correlation between score and true error.

4. RESULTS

4.1 Selection on Diffusion Policy Candidates

The DP baseline has a translation error of 0.117 [0.114, 0.120] and a rotation error of 40.01° [39.61, 40.42]. Table 2 reports every scorer on this row, and V-JEPA2-AC is the only one with a validated improvement.

- B2 reduces translation error by 5.98% [5.43, 6.47] and rotation error by 0.96°.
- B3 reduces them by 5.37% and 0.85°, so the scorer finds better candidates elsewhere in the pool and does not only avoid the policy's bad choices.
- Against candidate 0, B2 wins in 49.88% of windows, ties in 23.17%, and loses in 26.95%. The ties arise because B2 keeps candidate 0 whenever the scorer ranks it first.

Table 2. DP row, change against candidate 0, pooled across 3 seeds [95% CI]. Negative is an improvement. † held out of the headline comparison (Section 4.4). ‡ single seed.

Scorer	Br.	Translation (%)	Rotation (°)
V-JEPA2-AC	B2	−5.98 [−6.47, −5.43]	−0.96 [−1.07, −0.83]
	B3	−5.37 [−5.77, −4.95]	−0.85 [−0.95, −0.76]
V-JEPA2 no-AC	B2	0.00 [0.00, 0.00]	0.00 [0.00, 0.00]
PointWorld proxy	B2	+0.27 [−0.04, +0.58]	−0.39 [−0.49, −0.28]
DexWM‡	B2	+0.57 [+0.12, +1.02]	−0.05 [−0.19, +0.08]
LeWM rollout†	B2	−10.09 [−10.34, −9.84]	−0.83 [−0.90, −0.76]

Seed-to-seed spread is small, with per-seed B2 reductions of 6.41%, 6.09%, and 5.46% and a baseline between 0.114 and 0.119. Why V-JEPA2-AC is the main positive result, and why LeWM's larger gain and the other scorers are not read the same way, is set out in Section 4.4.

4.2 How Much Is Attainable

The oracle ceiling for the same five candidates is 16.51% under B2 and 14.68% under B3. V-JEPA2-AC recovers 36.2% and 36.6% of these, a little over a third of what the pool allows. Its per-window Spearman correlation between score and true translation error is 0.164 [0.155, 0.170], modest but clearly above zero and flat across pool sizes. The remaining error belongs to the candidates the policy generates, and re-ranking cannot reach below it (Figure 3).

The achievable gain grows with pool size and does not plateau. V-JEPA2-AC's B2 translation reduction was 6.0% at $K = 5$, 7.2% at $K = 8$, and 9.1% at $K = 20$, so the pool of five understates what selection can deliver. The pool size is a known compute constraint, and a larger pool might lift the attainable ceiling further. Figure 4 summarises which routes meet the conditions.

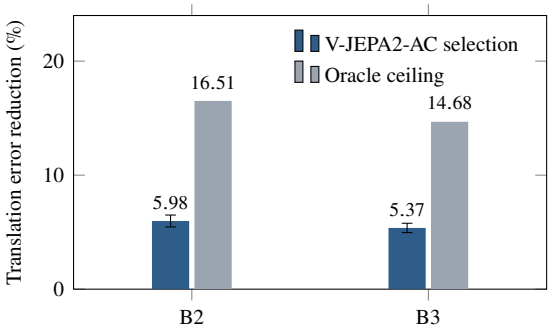


Figure 3. Error reduction achieved by V-JEPA2-AC against the oracle ceiling for the same five candidates. Whiskers are 95% CIs.

Route	Diverse pool	Action information	In-support candidates	Observed effect
DP + V-JEPA2-AC	Yes	Yes	Yes	−5.98% translation
DP + V-JEPA2 no-AC	Yes	No	Yes	0.00%
DP + PointWorld proxy	Yes	Partly	Yes	+0.27% n.s.
DINO-WM, JEPA-WM (adapter)	Not shown	Partly	No	Latent score only
DP + LeWM	Yes	Not shown	Yes	Gain not validated
ACT + CEMP + V-JEPA2-AC	Partly	Yes	Yes	−0.58%, circular
DiWA C2, external selection	Yes	Yes	Not shown	Small, mixed
DiWA C3b, imagined PPO	Yes	Yes	By construction	+14 to +40 pp

Figure 4. Evidence for the three conditions across routes. "Partly" marks a weakened or search-restored condition, and "Not shown" marks a condition this study could not establish for that route.

4.3 Where Selection Fails

An aggregate gain can hide a local failure, so windows were sorted by how accurate candidate 0 already was and split into five equal groups. Selection helps in four of them and hurts in one. In the easiest fifth, where the policy's own choice is already close to the best it can produce, the world model's pick is worse than keeping candidate 0. Translation error rises by 5.81% [4.49, 7.29] in that group, at all three seeds. Capture is the share of the oracle ceiling recovered, so an error increase appears as a negative value, −33.1%. In the other four groups capture climbs from 24.9% to about 50% as candidate 0 becomes less accurate (Figure 5).

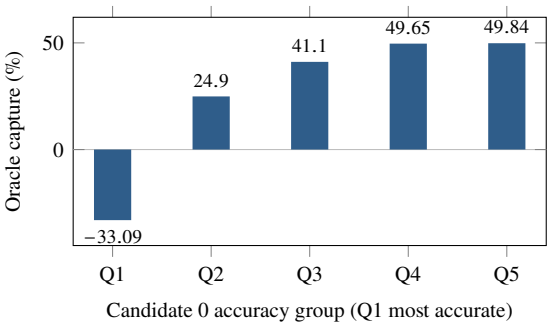


Figure 5. Share of the oracle ceiling recovered by V-JEPA2-AC, by how accurate candidate 0 already was. A negative value means selection made the result worse than candidate 0. Pooled across 3 seeds, pool of eight candidates, so not directly comparable with the pool-of-five figures.

A natural remedy is to abstain when the scorer is unsure. This did not help, because the gap between the top two scores has no usable relationship with baseline error (correlation −0.002), so the scorer cannot tell when it is in the harmful group. A windowing placed around hand-state changes removes most of the reversal (margin −0.54%), so the failure depends on how easy windows are sampled and is not fixed to the scorer.

4.4 The Remaining Scorers

The zero-action ablation returns the baseline exactly in B2, because it assigns tied scores and never overrides candidate 0. This confirms that the action-conditioned head, not the pretrained visual features alone, carries the signal. The PointWorld proxy leaves translation indistinguishable from zero while giving a small rotation gain, consistent with its design as a control. DexWM worsens translation error in both branches. This is a valid result for the fixed-budget predictor evaluated here, because the published pretrained predictor was not released and the original training scale was outside the local budget.

LeWM posts the largest numerical gain, a 10.09% translation reduction, but it fails the ranking-validity gate. Its Spearman correlation is −0.51 in B2 and −0.52 in B3 on the full test set, and a control that disables its dynamics model gives a nearly identical gain of 10.24%. Something other than the scorer's stated logic is therefore driving the improvement, and LeWM is reported as an unvalidated result and held out of the headline comparison. The routes are summarised below.

- **V-JEPA2-AC.** Frozen ViT-L with a local action head. A validated −5.98% translation change, because the action-conditioned head carries the signal (no-AC ablation 0.00%).
- **DexWM.** DINOv2 encoder with a predictor trained at a fixed budget. +0.57% (worse) on one seed, with a likely explanation that the predictor is under-trained relative to the unreleased published recipe.
- **LeWM.** Compact encoder trained from scratch. −10.09% with a rank correlation of −0.51, so the gain is not attributable to its dynamics.
- **DINO-WM and JEPA-WM.** Official checkpoints through a train-only linear action adapter. The latent score improves while the trajectory metric stays flat or worse, likely because the adapter loses information compressing 18-D actions into a 7-D interface.
- **PointWorld proxy.** V-JEPA2-AC pathway with weakened action input. +0.27% and not significant, a control that separates action information from visual quality.

4.5 ACT Results

ACT's pool comes from CEMP (Section 3.2), so these results describe the planner and world model together and not the scorer alone. V-JEPA2-AC gives a small improvement of 0.58% [0.14, 0.86] under B2 and 0.59% under B3, in the same direction as the DP row and much smaller, while the controls stay at or above zero in translation. The small size is expected, since a pool shaped by the scorer leaves it little to choose between.

4.6 Reproduction on CALVIN

As a secondary study, the DiWA protocol is reproduced on four CALVIN tasks using the authors' own checkpoints,

pooled across the same three seeds [5, 21]. C1 is DiWA's base policy alone. C2 adds this paper's external selection, using DiWA's own world model and reward classifier. C3b is the policy after DiWA's imagined-PPO finetuning (Figure 6).

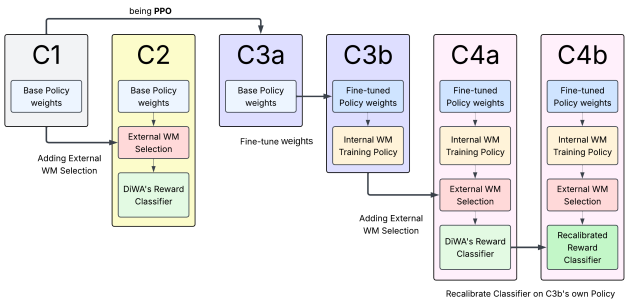


Figure 6. The six DiWA conditions. C1 is the base policy. C2 adds external selection. C3a and C3b finetune the weights with imagined PPO. C4a and C4b add selection on top of the finetuned policy, and C4b recalibrates the reward classifier on that policy's own rollouts.

Table 3 reports task success. External selection moves three tasks backward by 2.3 to 3.1 points and one forward by 6.1 points, with overlapping intervals throughout. The reward classifier selects a non-first candidate on 79% to 81% of decisions, so the pool offers a real choice. Finetuning moves the same starting point by 14 to 40 points on every task. C4a and C4b, which add selection on the finetuned policy, were statistically inconclusive, and both reuse the world model that already shaped the policy, so little new information should be expected from a second consultation.

Table 3. CALVIN task success (%), pooled across 3 seeds, roughly 100 episodes per task. C3b is DiWA's trainer-loop reading.

Task	C1	C2	C3b
close_drawer	59.8	57.5	93.1
turn_on_led	52.9	50.6	90.8
move_slider_left	57.3	54.2	82.3
turn_on_lightbulb	67.7	73.7	87.9

5. DISCUSSION

5.1 Three Conditions

Read across all routes, selection helps when three conditions hold together. Figure 4 sets the evidence out. The pattern is offered as what this study's experiments support, not as a general law.

A diverse pool is needed because a scorer cannot improve on a pool with one distinct answer, as the ACT collapse showed (Section 3.2). Action-conditioned information is needed because a scorer that does not see the action cannot discriminate between actions, as the no-AC ablation showed with an exact zero. Compatibility with training support matters because a model can improve its own latent score while the trajectory metric stays flat, as the adapter pilots showed (Section 4.4). These routes failed in different ways, and the conditions give one frame for reading them together.

5.2 Selection Versus Finetuning

The routes place the world model in three different roles. In DiWA it is a training-time component, and PPO writes its judgement into the policy weights. In the DP row it is an inference-time external ranker with no influence on the policy, and in CEMP it also searches, perturbing a frozen output and ranking the perturbations.

DiWA's finetuning meets all three conditions by construction. Its reward classifier is the signal PPO optimises against, so the policy is pushed toward what the classifier already rewards. External selection has no such guarantee, because the scorer and the policy are shaped independently. The order-of-magnitude gap on CALVIN follows. The EgoDex setting has no interactive environment to imagine rollouts in, since it consists of pre-recorded human video, so external selection is what that data supports. The CALVIN result does not show that selection is the stronger mechanism in general.

5.3 Limitations

Several limits qualify these conclusions. All primary results use one dataset, and the V-JEPA2-AC head is trained on EgoDex actions, so transfer to another action distribution should not be assumed. The benchmark is offline and trajectory-level, and it does not measure task success or physical execution, apart from the CALVIN reproduction. The primary claim rests on one policy family and one world-model pairing. Exposure is matched locally, but externally pretrained backbones carry training the budget does not offset. DexWM is single-seed, and the LeWM question is unresolved. Selection also reverses on the easiest windows, and a confidence gate does not fix this. The CALVIN comparison uses about 100 episodes per task with overlapping intervals for selection. The three conditions are inferred across routes and not isolated by ablation.

6. CONCLUSIONS AND FUTURE WORK

A frozen world model used only to re-rank a policy's own candidates can improve selected hand motion, but under the tested conditions this holds for one pairing. DP candidates scored by V-JEPA2-AC reduce translation error by 5.98% and recover about 36% of the attainable gain, without retraining. The four other routes did not reproduce this, and each has a likely explanation or limitation. On CALVIN, weight-level finetuning outperforms external selection by roughly an order of magnitude, and it does so because it meets the three conditions by construction. The evaluation protocol itself is part of the contribution, because fixed windows, frozen pools, and blocked goal access are what allowed earlier, stronger-looking results to be identified as shortcuts and corrected. The three objectives are met by the benchmark, the comparison, and the account of when selection helps.

Future work includes the following.

- Extending the protocol to closed-loop execution.
- Testing a finetuned policy against a second, independently trained world model as the external selector.
- Finetuning DINO-WM and JEPA-WM natively on 18-dimensional actions.
- Evaluating DexWM at its published scale once a checkpoint is available.
- Giving ACT an architectural source of diversity, such as discrete latents, so that it no longer needs a scorer-shaped pool.
- Testing larger candidate pools, since the achievable gain did not plateau by $K = 20$.

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